Course Code	Course Title	L	T	P	C
MACSE636	Cyber Security	3	0	0	3
Pre-requisite	Nil	**Syllabus Version**			
		1			

Course Objectives

To introduce the fundamental principles of cybersecurity
To impart comprehensive knowledge and practical skills in cryptographic techniques
To develop the ability to identify, understand, and mitigate various types of cybercrimes

Course Outcomes

Understand and apply core principles of cybersecurity, including data integrity and authentication mechanisms
Demonstrate knowledge of classical and modern cryptographic techniques and their practical applications.
Identify, analyze, and evaluate various types of cybercrimes, threats, and attack vectors.
Develop, implement, and assess effective cybersecurity policies and strategies for organizations.

Module:1	**Overview of Cyber Security and Risk Management**	**7 hours**
Key principles: Confidentiality, Integrity, and Availability (CIA), Threats and vulnerabilities: attacks, defense and weaknesses-Risk analysis and assessment: Identifying and evaluating potential risks		
Module:2	**Cryptography Techniques**	**9 hours**
Encryption methods: Classical techniques, Symmetric cryptography- block ciphers: DES-AES, DES, stream ciphers: RC4, and Asymmetric cryptography: RSA, ElGamal, Elliptic Curve, Key Distribution		
Module:3	**Integrity and Authentication**	**9 hours**
Hash functions: Message Digest (MD5), Secure Hash Algorithm (SHA-256), Message Authentication, Message Authentication Code (MAC), Digital Signature Algorithm: RSA ElGamal based, HMAC		
Module:4	**Cybercrimes and Cyber Offenses**	**9 hours**

Types of cybercrimes: Classification and characteristics, Attack planning and execution, social engineering tactics: Human-based and computer-based attacks: Cyber stalking and Cyber Cafe, phishing, password cracking, keyloggers, spyware, DoS/DDoS attacks, and SQL injection, reconnaissance and disruption methods, malicious codes		
Module:5	**Cyber Security Policy Development and Enforcement**	**9 hours**
Identifying Organizational policy needs: Determining requirements for security policies, Policy creation for Internet Use, Compliance and enforcement: Ensuring adherence to policies and procedures, Policy review and updates		
Module:6	**Contemporary Issues**	**2 hours**
	Total Lecture Hours:	**45 hours**

Text Book(s)

William Stallings, "**Cryptography and Network Security: Principles and Practice**", Pearson Education, 8th Edition, **2023**
Nina Godbole & Sunit Belapure, "**Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives**", Wiley, 1st Edition, **2011**
Scott Barman, "**Writing Information Security Policies**", New Riders Publishing, 1st Edition, **2002**

Reference Books

Brian Underdahl, "**Cybersecurity for Dummies**", Wiley Publishing Inc , 1st Edition, **2011**
Joseph Steinberg, Kevin Beaver, Ira Winkler, "**Cybersecurity All-in-One For Dummies**", John Wiley and Sons Inc, 1st Edition, **2023**
Behrouz A. Forouzan, Debdeep Mukhopadhyay, "**Cybersecurity All-in-One For Dummies**", Mcgraw Hill Education, 3rd Edition, **2015**

Mode of Evaluation : Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test

Recommended by Board of Studies :	23-05-2025
Approved by Academic Council : No. 78	12-06-2025